1.10%) for a 10 μg/m^3 increase in daily 24-hour mean $PM_{2.5}$ concentrations for the same day (Peng et al., 2009, page 960).

Note that Peng et al. (2009) considered a broader range of ICD-9 codes and estimated the risk of both cardiovascular events and cerebro- and peripheral vascular disease. For comparability to other studies, EPA decided to apply a baseline hospitalization rate for ICD-9 codes 390-409 and 411-429 when using this C-R function in quantifying impacts.

G.1.3.10 Sheppard (2003)

Sheppard et al. (1999) studied the relation between air pollution in Seattle and nonelderly (<65) hospital admissions for asthma from 1987 to 1994. They used air quality data for PM_{10}, $PM_{2.5}$, coarse $PM_{10-2.5}$, SO_2, ozone, and CO in a Poisson regression model with control for time trends, seasonal variations, and temperature-related weather effects. $PM_{2.5}$ levels were estimated from light scattering data. They found asthma hospital admissions associated with PM_{10}, $PM_{2.5}$, $PM_{10-2.5}$, CO, and ozone. They did not observe an association for SO_2. They found PM and CO to be jointly associated with asthma admissions. The best fitting co-pollutant models were found using ozone. However, ozone data was only available April through October, so they did not consider ozone further. For the remaining pollutants, the best fitting models included $PM_{2.5}$ and CO. Results for other co-pollutant models were not reported.

In response to concerns that the work by Sheppard et al. (1999) may be biased because of the Splus issue, Sheppard (2003) reanalyzed some of this work, in particular Sheppard reanalyzed the original study's $PM_{2.5}$ single pollutant model.

Hospital Admissions, Asthma (ICD-9 code 493)

The coefficient and standard error are based on the relative risk (1.04) and 95% confidence interval (1.01-1.06) for a 11.8 μg/m^3 increase in $PM_{2.5}$ in the 1-day lag GAM stringent model (Sheppard, 2003, pp. 228-229).

G.1.3.11 Zanobetti et al. (2009)

Zanobetti et al. (2009) examined the relationship between daily $PM_{2.5}$ levels and emergency hospital admissions for cardiovascular causes, myocardial infarction, congestive heart failure, respiratory disease and diabetes among 26 U.S. communities from 2000-2003. The authors used meta-regression to examine how this association was modified by season- and community-specific $PM_{2.5}$ composition while controlling for seasonal temperature as a substitute for ventilation. Overall, the authors found that $PM_{2.5}$ mass higher in Ni, As, and Cr as well as Br and organic carbon significantly increased its effects on hospital admissions. For a 10 μg/m^3 increase in 2-day averaged $PM_{2.5}$, the authors found a 1.89% (95% CI: 1.34-2.45) increase in cardiovascular disease admissions, a 2.25% (95% CI: 1.10-3.42) increase in myocardial infarction admissions, a 1.85% (95% CI: 1.19-2.51) increase in congestive heart failure admissions, a 2.74% (95% CI: 1.30-4.20) increase in diabetes admissions, and a 2.07% (95% CI: 1.20-2.95) increase in respiratory admissions. The relationship between $PM_{2.5}$ and cardiovascular admissions was significantly modified when the mass of $PM_{2.5}$ was high in Br, Cr, Ni, and sodium ions, while mass high in As, Cr, Mn, organic carbon, Ni and sodium ions modified the myocardial infarction relationship and mass high in As, organic carbon, and sulfate ions modified the diabetes admission rates.

Hospital Admissions, All Cardiovascular (ICD-9 codes 390-429)

In a single-pollutant model, the coefficient and standard error are estimated from the percent change in risk (1.89%) and 95% confidence interval (1.34%-2.45%) for a 10 μg/m^3 increase in 2-day averaged $PM_{2.5}$ (Zanobetti et al., 2009, Table 3).

Note that Zanobetti et al. (2009) report results for ICD codes 390-429. In the benefit analysis, avoided nonfatal heart attacks are estimated separately. In order to avoid double counting heart attack hospitalizations, we have excluded ICD code 410 from the baseline incidence rate used in this function.

Hospital Admissions, All Respiratory (ICD-9 codes 460-519)

In a single-pollutant model, the coefficient and standard error are estimated from the percent change in risk (2.07%) and 95% confidence interval (1.2% - 2.95%) for a 10 μg/m^3 increase in 2-day averaged $PM_{2.5}$ (Zanobetti et al., 2009, Table 3).

G.1.4 Emergency Room Visits

Table G-8 summarizes the additional health impacts functions used to estimate the relationship between $PM_{2.5}$ and emergency room visits. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-8. Additional Health Impact Functions for Particulate Matter and Emergency Room Visits

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Asthma	Glad et al.	2012	Pittsburgh, PA	0-99		D24HourMean	0.00392	0.00284	Logistic	All Races
Asthma	Mar et al.	2010	Greater Tacoma, WA	0-99		D24HourMean	0.0056	0.0021	Log-linear	
Emergency Room Visits, Asthma	Norris et al.	1999	Seattle, WA	0-17	NO_2, SO_2	D24HourMean	0.016527	0.004139	Log-linear	
Asthma	Slaughter et al.	2005	Spokane, WA	0-99		D24HourMean	0.0029	0.0027	Log-linear	

G.1.4.1 Glad et al. (2012)

Glad et al. (2012) examined the relationship between air pollution and emergency department visits for asthma (ICD-9 code 493) in six Pittsburgh, PA hospitals between 2002 and 2005. The study includes a total of 6,979 individuals with a primary discharge diagnosis of asthma. Using a case-crossover methodology, particulate matter with an aerodynamic diameter ≤2.5μm ($PM_{2.5}$) had an effect both on the total population on day 1 after exposure (1.036, $p < 0.05$), and on African Americans for day-1 lag (OR =1.055;

95% CI, 1.001–1.112), day-2 lag (OR=1.067; 95% CI, 1.015–1.122), day-3 lag (OR=1.053; 95% CI, 1.002–1.106), and the 6-day average (OR =1.088; 95% CI, 1.001–1.184). $PM_{2.5}$ had no significant effect on Caucasian Americans alone. The disparity in risk estimates by race may reflect differences in residential characteristics, exposure to ambient air pollution, or a differential effect of pollution by race.

Emergency Room Visits, Asthma (ICD-9 Code 493) – by Age and Racial Group

In a single pollutant model for patients of all ages and all races, the coefficient and standard error were estimated from the odds ratio (1.04) and 95% confidence interval (0.984 – 1.10) for a 10 μg/m^3 increase in 24-hour mean $PM_{2.5}$ concentration averaged over lags 0-6 days (Glad et al., 2012, Table 3).

G.1.4.2 Mar et al. (2010)

Mar et al. (2010) assessed the effect of particulate matter air pollution, including emissions from diesel generators, on emergency room visits for asthma in the greater Tacoma, Washington area from January 3, 1998, to May 30, 2002, using Poisson regression models. Health data were collected for individuals of all ages from 6 Tacoma hospitals. The authors also assessed the impacts of diesel generator use on emergency room visits for asthma from January 24, 2001, to June 2, 2001. Overall, the researchers found an association between daily $PM_{2.5}$ levels and emergency room visits for asthma at lag days 2 and 3, with a relative risk for lag day 2 of 1.04 (95% CI: 1.01-1.07) and a relative risk for lag day 3 of 1.03 (95% CI: 1.0-1.06). No significant association between emergency room visits for asthma and increased use of the diesel generators was observed.

Emergency Room Visits, Asthma (ICD-9 code not reported)

In a single-pollutant model, the coefficient and standard error are estimated from the relative risk (1.04) and 95% confidence interval (95% CI: 1.01-1.07) for a 7 μg/m^3 increase in daily 24-hour mean $PM_{2.5}$ at lag day 2 (Mar et al., 2010, Table 4).

G.1.4.3 Norris et al. (1999)

Norris et al. (1999) examined the relation between air pollution in Seattle and childhood (<18) hospital admissions for asthma from 1995 to 1996. The authors used air quality data for PM_{10}, light scattering (used to estimate fine PM), CO, SO_2, NO_2, and O_3 in a Poisson regression model with adjustments for day of the week, time trends, temperature, and dew point. They found significant associations between asthma ER visits and light scattering (converted to $PM_{2.5}$), PM_{10}, and CO. No association was found between O_3, NO_2, or SO_2 and asthma ER visits, although O_3 had a significant amount of missing data. In multipollutant models with either PM metric (light scattering or PM_{10}) and NO_2 and SO_2, the PM coefficients remained significant while the gaseous pollutants were not associated with increased asthma ER visits. The PM C-R functions are based on results of the single and multipollutant models reported.

Emergency Room Visits, Asthma

In a model with NO_2 and SO_2, the $PM_{2.5}$ coefficient and standard error are calculated from a relative risk of 1.17 (95% CI 1.08-1.26) for a 9.5 μg/m^3 increase in $PM_{2.5}$ (Norris et al., 1999, p.491).

G.1.4.4 Slaughter et al. (2005)

Slaughter et al. (2005) examined the short-term association of particulate matter (PM1, $PM_{2.5}$, PM_{10}, and PM_{10}-2.5) and carbon monoxide with hospital admissions and emergency room visits for respiratory and cardiac outcomes and mortality in Spokane, Washington, from January 1995 to June 2001 using a log-linear generalized linear model. The authors found no association between respiratory emergency room visits and any size fraction of PM, but there was a suggestive relationship between fine PM and respiratory effects when compared to coarse PM. No association between cardiac hospital admissions or mortality and any size fraction of PM or CO was observed at the 0- to 3-day lag. CO, on the other hand, was found to be associated with all respiratory emergency room visits and visits for asthma at the 3-day lag.

Emergency Room Visits, Asthma (ICD-9 code 493)

In a single-pollutant model, the coefficient and standard error are estimated from the relative risk (1.03) and 95% confidence interval (95% CI: 0.98-1.09) for a 10 μg/m^3 increase in daily 24-hour mean $PM_{2.5}$ at 1-day lag (Slaughter et al., 2005, Table 4).

G.1.5 Minor Effects

Table G-9 summarizes the additional health impacts functions used to estimate the relationship between $PM_{2.5}$ and minor effects. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-9. Additional Health Impact Functions for Particulate Matter and Minor Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form
Acute Bronchitis	Dockery et al.	1996	24 communities	8-12		Annual	0.027212	0.017096	Logistic
Work Loss Days	Ostro	1987	Nationwide	18-64		D24HourMean	0.004600	0.000360	Log-linear
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	Ozone	D24HourMean	0.007410	0.000700	Log-linear
Lower Respiratory Symptoms	Schwartz and Neas	2000	6 U.S. cities	7-14		D24HourMean	0.019012	0.006005	Logistic

G.1.5.1 Dockery et al. (1996)

Dockery et al. (1996) examined the relationship between PM and other pollutants on the reported rates of asthma, persistent wheeze, chronic cough, and bronchitis, in a